



**FEDERAL PUBLIC SERVICE COMMISSION**  
**COMPETITIVE EXAMINATION-2021**  
**FOR RECRUITMENT TO POSTS IN BS-17**  
**UNDER THE FEDERAL GOVERNMENT**

Roll Number

**GEOLOGY**

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| <b>TIME ALLOWED: THREE HOURS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>PART-I (MCQS) MAXIMUM MARKS = 20</b> |
| <b>PART-I(MCQS): MAXIMUM 30 MINUTES</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>PART-II MAXIMUM MARKS = 80</b>       |
| <b>NOTE:</b> (i) <b>Part-II</b> is to be attempted on the separate <b>Answer Book</b> .<br>(ii) Attempt <b>ONLY FOUR</b> questions from <b>PART-II</b> by selecting <b>TWO</b> questions from <b>EACH SECTION</b> . <b>ALL</b> questions carry <b>EQUAL</b> marks.<br>(iii) All the parts (if any) of each Question must be attempted at one place instead of at different places.<br>(iv) Write Q. No. in the Answer Book in accordance with Q. No. in the Q.Paper.<br>(v) No Page/Space be left blank between the answers. All the blank pages of Answer Book must be crossed.<br>(vi) Extra attempt of any question or any part of the question will not be considered. |                                         |

**PART – II**  
**SECTION – I**

- Q. No. 2.** Classify pyroxene group of minerals based on chemical composition. Also mention their physical and optical properties. (20)
- Q. No. 3.** Elaborate various laws of stratigraphy, use suitable sketches/examples in support of your answer. (20)
- Q. No. 4.** Elaborate the fundamental silica tetrahedral structure ( $\text{SiO}_4$ ). Give possibilities of  $\text{SiO}_4$  linkage to give rise to six silicate structural groups. (20)
- Q. No. 5.** Describe the test compositions of order Foraminiferida. Discuss their ecology and how it plays important role in Tertiary biostratigraphy. (20)

**SECTION-II**

- Q. No. 6.** Elaborate characteristics of effective trap-rocks. Describe principal types of hydrocarbon traps with examples. (20)
- Q. No. 7.** What are the possible causes of land sliding in the northern mountainous region of Pakistan and how it can be reduced? (20)
- Q. No. 8.** Write short notes on any TWO of the following: (10 each) (20)
- (a) Geothermal energy
  - (b) Eustatic curve
  - (c) Physical weathering

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